SENSORS AND ACTUATORS

Robotics can be equipped with various sensors that mimic or enhance human senses, such as cameras for vision, microphones for hearing, and tactile sensors for touch. These sensory inputs enable robots to perceive and interact with their environment, allowing them to perform tasks and navigate their surroundings.

Terminologies

- ***Sensor*** - a device that measures some attribute of the world. A device, such as a photoelectric cell, that receives and responds to a signal or stimulus.
    - **Active sensors** put energy into the environment to either change the energy or enhance it. **Active sensing** connotes a system using an effector to position a sensor to improve data gathering dynamically.
    - **Passive sensors** rely on the environment to provide the medium for observation. For example, a camera requires a certain amount of ambient light to produce a usable picture.
- **Transducer** - a device actuated by power from one system and supplies power to a second system, usually in another form. In addition, the sensor's mechanism, or element, transforms the energy associated with what is being measured into another form of energy.
- **Actuator** - A mechanism for moving or controlling something indirectly instead of by hand.

Types of Errors in Sensors

- **Random error**: This is caused by fluctuations in the environment or in the sensor itself. Random error can be reduced by taking multiple measurements and averaging them.
- **Systematic error**: This is caused by a consistent sensor or measurement system bias. Systematic errors can be corrected by calibrating the sensor or adjusting the measurement system.
- **Drift error**: This occurs when the sensor readings gradually shift over time, even when no external factors are changing.
- **Interference error**: This is caused by interference from other sources, such as electromagnetic interference (EMI) or crosstalk between sensors.
- **Nonlinearity error**: This occurs when the relationship between the sensor input and output is not linear.
- **Hysteresis error**: This occurs when the sensor readings are affected by previous measurements, causing a lag or delay in the response.

Sensor Characteristics

- **Resolution** - The minimum difference between two values that the sensor can measure.
- **Accuracy** - The uncertainty in a sensor measurement with respect to an absolute standard.
- **Sensitivity** - The smallest absolute change that a sensor can measure.
- **Linearity** - Whether the output produced by a sensor depends linearly on the input.
- **Precision** - The reproducibility of the sensor measurement.
- **Bandwidth** - The speed at which a sensor can provide measurements.
- **Dynamic range** - Under normal operation, this is the ratio between the limits of the lower and upper sensor inputs.

Types of Sensors:

- **Shaft Encoders** - measure the number of turns the drive motor has made; the shaft encoder works the same way an odometer on a car works. **Odometry** is the use of motion sensors to determine the robot's temporal change in position relative to some known position.
    - **Linear encoders** - measure straight-line motion. Sensor heads that attach to the moving piece of machinery run along guideways. Those sensors are linked to a scale inside the encoder that sends digital or analog signals to the control system.
        - **Accelerometers** can also provide dead reckoning information and often do much better than odometry.
    - **Rotary encoders** - measure rotational movement. They typically surround a rotating shaft, sensing and communicating changes in its angular motion.
        - **Gyroscopes -** a device used to measure or maintain orientation and angular velocity. It consists of a spinning wheel or disc mounted on a set of gimbals, allowing it to rotate freely in any direction.
- **Proximity Sensors** - directly measure the relative distance (range) between the sensor and objects in the environment. It generally connotes measuring short distances to an object, on the order of one meter, versus a range sensor, which senses much larger distances.
    - **Ultrasonic Sensor** - uses sonar to determine the distance from an object. This sensor utilizes acoustic energy to detect objects and measure distances from the sensor to the target objects.
    - **Motion Sensor** - is an electronic device that is used for motion detection. It contains an electronic sensor that quantifies motion and can either be integrated with or connected to other devices that alert a user of the presence of a moving object within the sensor's field of view.
    - **Hall Sensor** - is an omni-polar magnet sensor used to detect magnetic objects. It can be used for proximity switching, positioning, speed detection, and current sensing applications.
- **Light Sensors** - are used to detect light that creates a difference in voltage signal to provide feedback to the robot's system. The two common light sensors that are widely used in the field of robotics are photoresistors and photovoltaic cells.
    - **Infrared Sensors (IR)** - emit near-infrared energy and measure whether any significant amount of the IR light is returned. If an obstacle is present, it returns a binary signal.
    - **Photodiode** - is a reverse-biased semiconducting optical sensor that is biased against its easy flow direction so that the current is very low.
    - **Flame Sensor** - can be used to detect fire or other wavelengths at 760-1100 nm light. The detection angle is roughly 60°, and the sensor is particularly sensitive to the flame spectrum.
- **Temperature Sensor** - provides temperature feedback to a system controller to make decisions.
- **Gas Sensors** - are widely used to detect gas leakages of LPG, Butane, Propane, Methane, Alcohol, Hydrogen, and smoke in houses and factories.
- **Joystick Module** - simply connects to two analog outputs based on the commands: X and Y control. It is made up of two potentiometers and a press button, which provides a solution to move robots to a target directly.
- **Force sensor** - is usually used to measure external mechanical force input, such as in the form of a load, pressure, or tension.
    - **Tactile sensors** - are generally used to mimic the sense of touch. Usually, tactile sensors are expected to measure small variations in force or pressure with high sensitivity.

  - **Bump Sensors** - consist of two layers forming a protruding ring around the robot. Contact with an object causes the two layers to touch, creating an electrical signal.
  - **Digital Tilt Sensor** - is a digital tilt switch in which there are two modules: mercury-based and ball-based.
  - **Capacitive Touch Sensor** - When any object with capacitive characteristics comes close to a capacitive touch sensor, it acts as another capacitor due to its dielectric nature.
  - **Vibration Sensor** - is a digital plug that plays sensor blocks. It is made up of a spring and a conductive material passing through the center of the spring.
- **Computer Vision** - refers to processing data from any modality that uses the electromagnetic spectrum to produce an image.
  - An **image** - is a way of representing data in a picture-like format where there is a direct physical correspondence to the scene being imaged. A different technique, **color histogramming**, can be used to identify a region with several colors.
    - **CMOS sensor** - The architecture of CMOS sensor arrays is like that of standard memories in which individual pixels can be randomly addressed and read out.
    - **Charge-coupled Devices (CCDs)** - In CCD technology, charges must be transferred from one pixel to the next until they can finally be read out at an array boundary.
  - **Line-tracking Sensor** - can detect white lines in black and black lines in white via the TTL signal.
  - **Color Sensor** - is a complete color detector that can detect and measure a nearly limitless range of visible colors. The applications include test strip reading, sorting color, ambient light sensing/calibration, etc.
  - **RGB-D sensors** - are unconventional visual sensors that can simultaneously obtain a visible image (RGB image) and a depth map of the same scene. They have been very popular in the robotics community for real-time image processing, robot localization, and obstacle avoidance.
- **Sound Sensor** - is used in detecting loud sounds in an ambient environment. It uses an electret condenser microphone to convert sound energy into an electrical signal, just like how a speaker operates.
  - **Voice recognition** - is a technology through which sounds, words, and phrases spoken by humans are converted into electrical signals.
- **Radio-frequency identification (RFID)** - This technology is based on the response of a **tag** to radio frequency signals.
  - The **tag** consists of an integrated circuit and an antenna, which provides identity to an **RFID reader**.
- **Biometric sensors** - Methods for biometric authentication include iris scans, fingerprint sensors, and face recognition.
- **Laser sensors** - can be utilized in several applications related to positioning. It is a remote sensing technology for distance measurement that involves transmitting a laser beam toward the target and analyzing the reflected light.
  - **LIDAR Light Detection And Ranging (LIDAR)** - has many applications in robotics, including object detection, obstacle avoidance, mapping, and 3D motion capture.
- **Global Positioning System** - works by receiving signals from satellites orbiting the Earth. The receiver calculates its position relative to at least four GPS satellites in terms of latitude, longitude, altitude, and change in time.

Actuators

It is a device or mechanism used to control or move a system or a component.

- **Motors -** these are commonly used to control the movement of robot joints, wheels, grippers, and other parts. They can provide precise and accurate control over motion, allowing robots to perform tasks with high accuracy and repeatability.
    - **Servo Motor** - is controlled with an electric signal, either analog or digital, which determines the amount of movement. It provides control of position, speed, and torque.
        - The ***speed of a DC motor*** is directly proportional to the supply voltage with a constant load.
        - An ***AC motor speed*** is determined by the frequency of the applied voltage and the number of magnetic poles.
    - **DC servo motors** - are commutated mechanically with brushes, using a commutator, or electronically without brushes.
        - **Brushed motors** - have two main parts: a rotor (the rotating part) and a stator (the stationary part). The rotor consists of a shaft with a permanent magnet, and the stator contains an electromagnet. When a current is applied to the stator, it creates a magnetic field that interacts with the rotor's magnetic field, causing the rotor to turn.
        - **Brushless motors** - do not have brushes or commutators. Instead, they have a stationary part called a stator that contains multiple electromagnets and a rotating part called a rotor that contains permanent magnets. A controller applies electrical currents to the electromagnets in the stator in a specific sequence, causing the rotor to turn.
    - **Stepper motor -** is a brushless synchronous DC motor that features precise discrete angular motions. This motor is designed to break up a single complete rotation into several much smaller and essentially equal part rotations.
    - **Linear motor -** operates on the same principle as an electric motor but provides linear motion. Unlike a rotary machine, a linear motor moves the object in a straight line or along a curved track.
- **Hydraulic actuators** - are driven by the pressure of the hydraulic fluid. It consists of a cylinder, piston, spring, hydraulic supply and return line, and stem. As such, they can be used in construction machinery and other heavy-duty equipment.
- **Pneumatic actuators** - have been known for being highly reliable, efficient, and safe sources of motion control. These actuators are driven by pressurized air that can convert energy in the form of compressed air into linear or rotary mechanical motion.

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